Diffusion Language Models Are Natively Length-Aware

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Abstract

Unlike autoregressive language models, which terminate variable-length generation upon predicting an End-of-Sequence (`EoS`) token, Diffusion Language Models (DLMs) operate over a fixed maximum-length context window for a predetermined number of denoising steps. However, this process is independent of the required response length, resulting in computational waste for the majority of short responses common in reasoning and chat tasks. To address this problem, we conjecture that the latent prompt representation contains sufficient information to estimate the required output length. We provide empirical evidence for this phenomenon and propose a zero-shot mechanism to dynamically crop the context window before generation begins, leading to fewer diffusion steps and substantial computational savings. We evaluate our approach on four benchmarks with diverse tasks—`GSM8K` (reasoning), `HumanEval` (code generation), `IfEval` (instruction following), and `LongFormQA` (question answering)—revealing massive efficiency gains at minimal performance impact. We report significant reductions in FLOPs across all tasks, with no statistically significant performance degradation, and significant performance improvements in 2 out of 4 tasks.

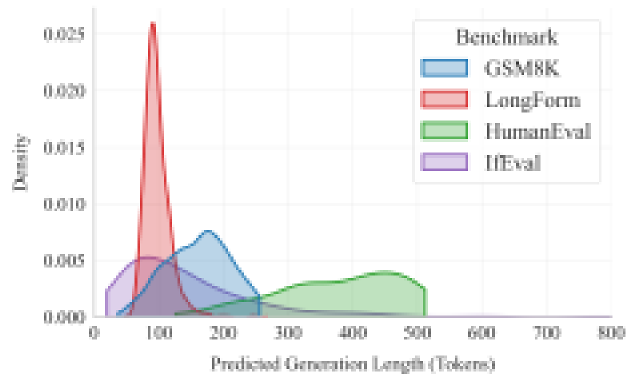

Figure 1. **Predicted Length Distributions.** Our SMARTCROP ($\tau = 0.9$) method successfully predicts task-specific output lengths across four benchmark datasets. The abrupt truncations observed in certain distributions correspond to context length constraints (refer to Section 4 for details).

1. Introduction

Language generation with Large Language Models (LLMs) has been dominated by autoregressive models, which generate text sequentially, predicting one token at a time (Zou et al., 2023). While proven successful, this sequential mechanism inherently limits inference speed and brings various disadvantages, such as the inability to perform global refinements during the generation process and error propagation due to "wrong" early tokens. Recently, Diffusion Language Models (DLMs, Austin et al., 2021; Lou et al., 2024) have emerged as a promising alternative, offering the potential for accelerated, non-autoregressive generation through iterative denoising DLMs operate by progressively unmasking tokens on a fixed-length canvas. This process is initialized with a prompt, while the remainder of the maximum context window is filled with placeholder mask tokens. In each denoising step, the model predicts logits for the entire masked sequence and unmasks a subset of tokens, typically those with the highest confidence. Unmasked tokens are kept and the process repeated. While this allows for flexible, parallel sequence generation, the requirement of a fixed canvas length remains a significant bottleneck, as the dimensions must be defined *a priori* based on heuristics or domain knowledge. To support variable-length outputs, current approaches often rely on padding with special End-of-Sequence (`EoS`) tokens to prevent unmasking beyond a certain point (Nie et al., 2025). However, this introduces substantial computational waste: the model must still process the entire context window during every forward pass, regardless of the actual output length.

In this work, we conjecture that DLMs implicitly encode the required output length within their latent representation of the prompt. In other words, DLMs encode an expectation about how many answer tokens are needed, depending on the task or question they are prompted with. While these

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